homework-3

2026-09-17

Dylan Smith Homework 3

Question 1: Multiple Linear Regression on Carseats

(a) Full Model: Sales ~ Population + Income + US

```
carseats_full <- lm(Sales ~ Population + Income + US, data = Carseats)
summary(carseats_full)
```

```
##
## Call:
## lm(formula = Sales ~ Population + Income + US, data = Carseats)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.3916 -1.8389 -0.0199  1.7841  8.6010
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 5.7134361  0.4657440  12.267  < 2e-16 ***
## Population  0.0007997  0.0009395   0.851  0.39519
## Income      0.0139000  0.0049584   2.803  0.00531 **
## USYes       0.9562235  0.2901542   3.296  0.00107 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.76 on 396 degrees of freedom
## Multiple R-squared:  0.05177,    Adjusted R-squared:  0.04458
## F-statistic: 7.206 on 3 and 396 DF,  p-value: 0.0001016
```

(b) Coefficient Interpretations and US Reference Level

The intercept predicts what the sales will be when all of the predictor variables are equal to zero. In this case, the intercept is 5.7134361. The coefficient for the predictor variable Population is 0.0007997. This means that if we hold the other two predictor variables Income and US constant, an increase of one in Population will increase Sales by 0.0007997. The coefficient for the predictor variable Income is 0.0139000. This means taht if we hold Population and US constant, an increase of one in Income will increase Sales by 0.0139000. The coefficient for US is 0.9562235. US is a categorical variable, so the coefficient compares the value of US = Yes to the value of US = No. This means that if we hold Population and Income constant, a store located in the US will result in an increase of 0.9562235 in Sales.

(c) Fitted Regression Equation and Indicator Definitions

The fitted regression equation is

$$\widehat{\text{Sales}} = 5.7134361 + 0.0007997\,\text{Population} + 0.0139000\,\text{Income} + 0.9562235\,I_{\text{US}},$$

where the dummy variable for US is defined as

$$I_{\text{US}} = \begin{cases} 1, & \text{if US = Yes (the store is in the United States),} \\ 0, & \text{if US = No (the store is outside the United States).} \end{cases}$$

$\widehat{\text{Sales}}$ represents predicted sales.

(d) Significant Predictors at $\alpha = 0.05$

At $\alpha = 0.05$ reject the null hypothesis for the predictors Income and US. Their p-values are 0.00531 and 0.00107 respectively which are both less than 0.05.

(e) Reduced Model Using Significant Predictors

```
carseats_reduced <- lm(Sales ~ Income + US, data = Carseats)
summary(carseats_reduced)
```

```
##
## Call:
## lm(formula = Sales ~ Income + US, data = Carseats)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.5789 -1.8654  0.0172  1.8241  8.5487
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 5.919300   0.397882  14.877  < 2e-16 ***
## Income      0.013844   0.004956   2.793 0.005472 **
## USYes       0.971417   0.289504   3.355 0.000869 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.759 on 397 degrees of freedom
## Multiple R-squared:  0.05003,    Adjusted R-squared:  0.04524
## F-statistic: 10.45 on 2 and 397 DF,  p-value: 3.762e-05
```

(f) Full vs. Reduced Model: R^2, Adjusted R^2, RSE, F-Statistic, and Model Preference

The full models has a slightly higher R^2 value of 0.05177 compared to the reduced model which has a value of 0.05003. This means that removing the predictor variable Population results in a very slight loss in the model's explanatory ability for predicting Sales.

For the Adjusted R^2 value, the reduced model has a slightly larger value of 0.04524 compared to the full model's value of 0.04458. This very small difference in both the Adjusted and Multiple R^2 values in both models suggests that the predictor variable Population is not statistically significant.

The RSE for the full model is 2.760 and the RSE for the reduced model is 2.759. This very small difference between the RSE values continues to suggest that Population is not statistically significant.

For the full model, the F-statistic is F = 7.206 and for the reduced model the F-statistic is F = 10.45. Both models are shown to be statistically significant, meaning that they each contain at least one predictor variable associated with Sales.

Overall, the reduced model is preferred. Removing the predictor variable Population results in almost no change in the Multiple R^2, the Adjusted R^2, or the RSE. The reduced model is also still statistically significant after removing Population, as shown by the F-statistic.

(g) Reduced Model: 95% Confidence Intervals for All Coefficients

```
confint(carseats_reduced, level = 0.95)
```

```
##                   2.5 %     97.5 %
## (Intercept) 5.137081100 6.70151976
## Income      0.004099784 0.02358724
## USYes       0.402263922 1.54056948
```

The 95% confidence intervals for the reduced model's regression coefficients are:

- Intercept (β_0): (5.137081100, 6.70151976).
- Income (β_{Income}): (0.004099784, 0.02358724).
- USYes (β_{USYes}): (0.402263922, 1.54056948).

The USYes coefficient compares US stores to non-US stores, holding Income constant. Neither the Income nor the USYes confidence interval includes zero, consistent with both predictors being statistically significant at the 5% level in the reduced model.

Question 2: Simulation Study & Simple Linear Regression

(a) Generate x: 150 Standard Normal Observations, with Seed 2025

```
set.seed(2025)
x <- rnorm(n = 150, mean = 0, sd = 1)
```

(b) Generate Error Terms: Mean 0 and Variance 0.36

```
epsilon <- rnorm(n = 150, mean = 0, sd = 0.6)
```

(c) Generate y: Vector Length and True Population Parameters

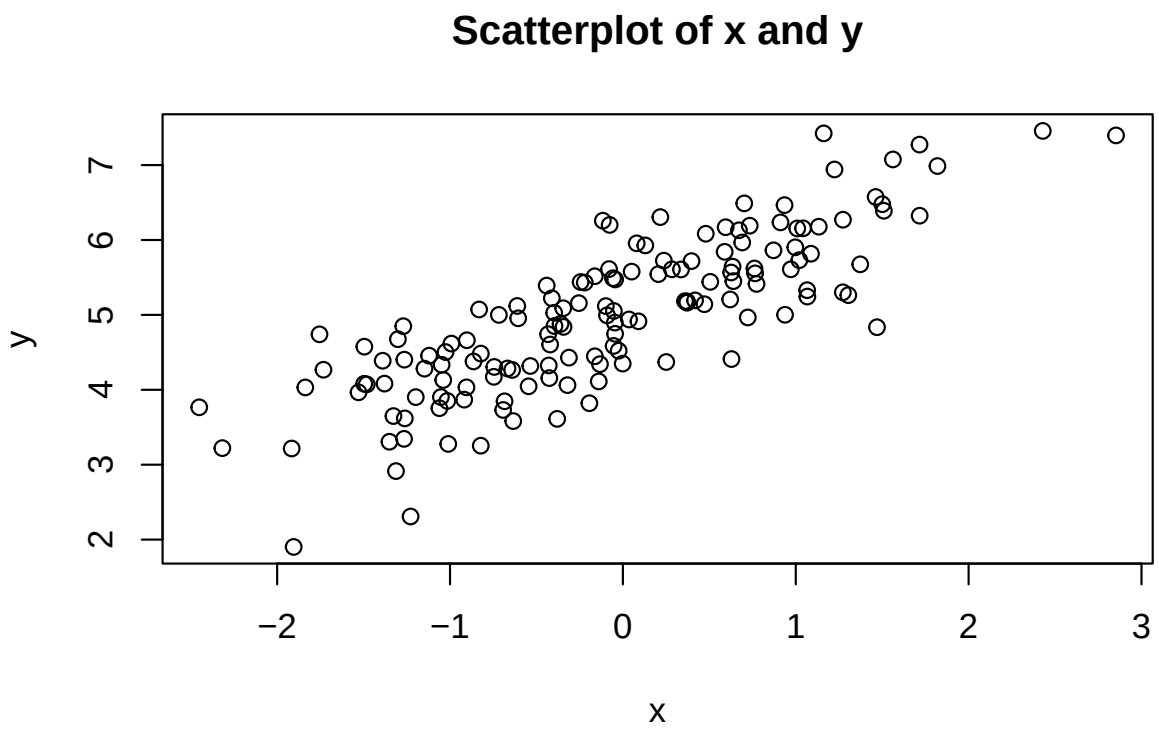
```r
y <- 5 + 0.8 * x + epsilon
length(y)
```

```
## [1] 150
```

The length of vector y is 150. The true population intercept is $\beta_0 = 5$, and the slope is $\beta_1 = 0.8$.

(d) Scatterplot of x and y: Patterns, Direction, and Spread

```r
plot(x, y,
     main = "Scatterplot of x and y",
     xlab = "x", ylab = "y")
```



The scatterplot shows a strong positive, approximately linear relationship: as x increases, y tends to increase. The points show moderate vertical spread around the linear trend, likely due to random variation. There is no obvious curvature or clear change in the spread as x increases.

(e) Least-Squares Model: Estimated vs. True Parameters

```r
simple_regression_model <- lm(y ~ x)
summary(simple_regression_model)
```

```
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.70753 -0.33985  0.04301  0.40568  1.36783
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.06374    0.04805  105.39   <2e-16 ***
## x            0.85355    0.04846   17.61   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5861 on 148 degrees of freedom
## Multiple R-squared:  0.677,  Adjusted R-squared:  0.6748
## F-statistic: 310.2 on 1 and 148 DF,  p-value: < 2.2e-16
```

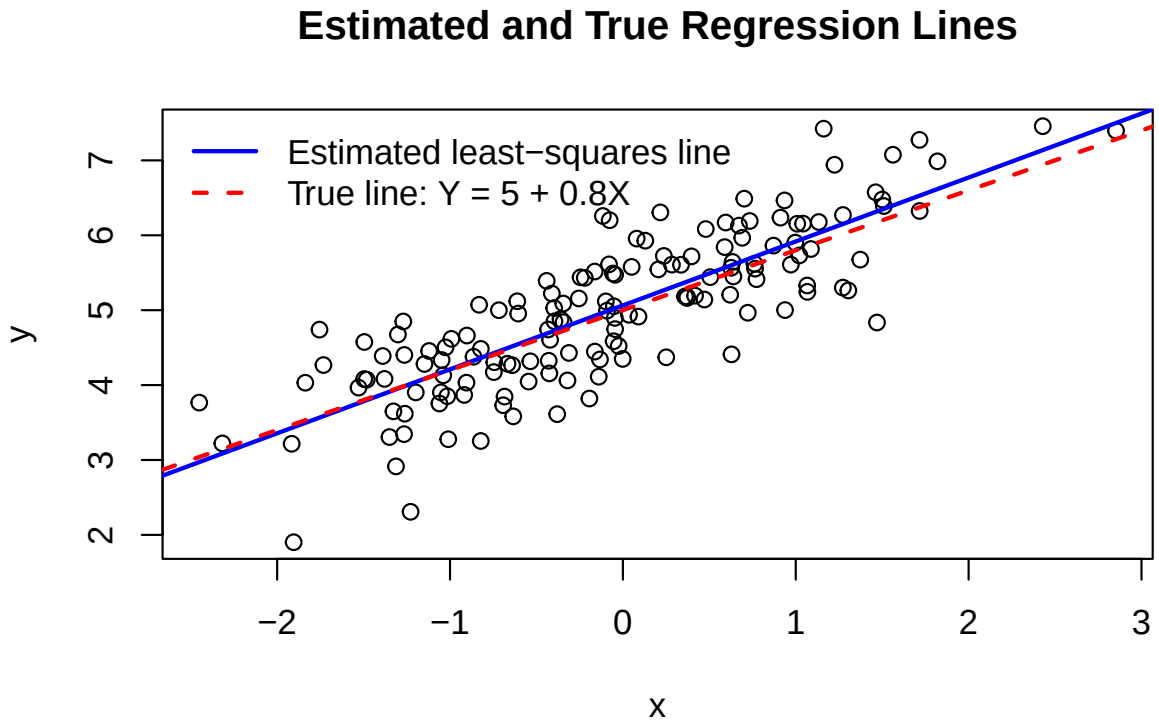
The estimated intercept is $\hat{\beta}_0 = 5.06374$, and the estimated slope is $\hat{\beta}_1 = 0.85355$. Thus, the fitted regression equation is

$$\hat{y} = 5.06374 + 0.85355x.$$

Both estimates are close to the true population parameters, $\beta_0 = 5$ and $\beta_1 = 0.8$. The differences between the estimates and the true population parameters reflect sampling variability from the randomly generated data.

(f) Scatterplot with Estimated and True Regression Lines and Legend

```r
plot(x, y,
     main = "Estimated and True Regression Lines",
     xlab = "x", ylab = "y")
abline(simple_regression_model, col = "blue", lty = 1, lwd = 2)
abline(a = 5, b = 0.8, col = "red", lty = 2, lwd = 2)
legend("topleft",
       legend = c("Estimated least-squares line", "True line: Y = 5 + 0.8X"),
       col = c("blue", "red"), lty = c(1, 2), lwd = 2,
       bty = "n")
```



(g) Quadratic Model: Evidence from the x^2 Term's p-Value

```r
quadratic_model <- lm(y ~ x + I(x^2))
summary(quadratic_model)
```

```
##
## Call:
## lm(formula = y ~ x + I(x^2))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7205 -0.3349  0.0558  0.4018  1.3576
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.03825    0.06093  82.686   <2e-16 ***
## x            0.85412    0.04855  17.591   <2e-16 ***
## I(x^2)       0.02599    0.03808   0.682    0.496
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5871 on 147 degrees of freedom
## Multiple R-squared:  0.678,  Adjusted R-squared:  0.6737
## F-statistic: 154.8 on 2 and 147 DF,  p-value: < 2.2e-16
```

The fitted quadratic model is

$$\hat{y} = 5.03825 + 0.85412x + 0.02599x^2.$$

To determine whether the quadratic term improves the model fit, we test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$ in the model $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$. The p-value for the x^2 term is 0.496. Since this is greater than the significance level of 0.05 that we have been using previously, we do not reject H_0. There is insufficient statistical evidence that adding the quadratic term improves the fit over the simple linear model.

(h) Noise Reduction Analysis

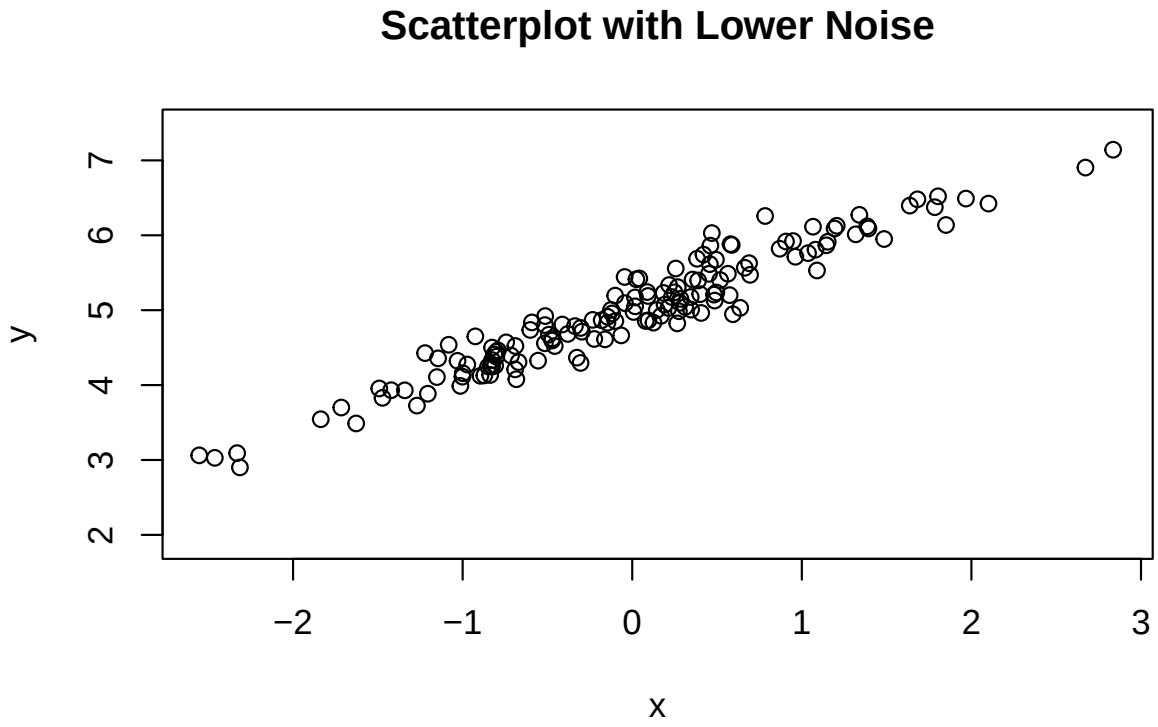
```
x_low_noise <- rnorm(n = 150, mean = 0, sd = 1)
epsilon_low_noise <- rnorm(n = 150, mean = 0, sd = 0.2)
y_low_noise <- 5 + 0.8 * x_low_noise + epsilon_low_noise
length(y_low_noise)
```

Repeat (a)–(c): Generate Data with Lower Error Variance

```
## [1] 150
```

```
plot(x_low_noise, y_low_noise,
     main = "Scatterplot with Lower Noise",
     xlab = "x", ylab = "y",
     xlim = range(x, x_low_noise), ylim = range(y, y_low_noise))
```

Repeat (d): Scatterplot with Lower Noise




```r
low_noise_model <- lm(y_low_noise ~ x_low_noise)
summary(low_noise_model)
```

Repeat (e): Fit the Lower-Noise Model and Compare Parameters

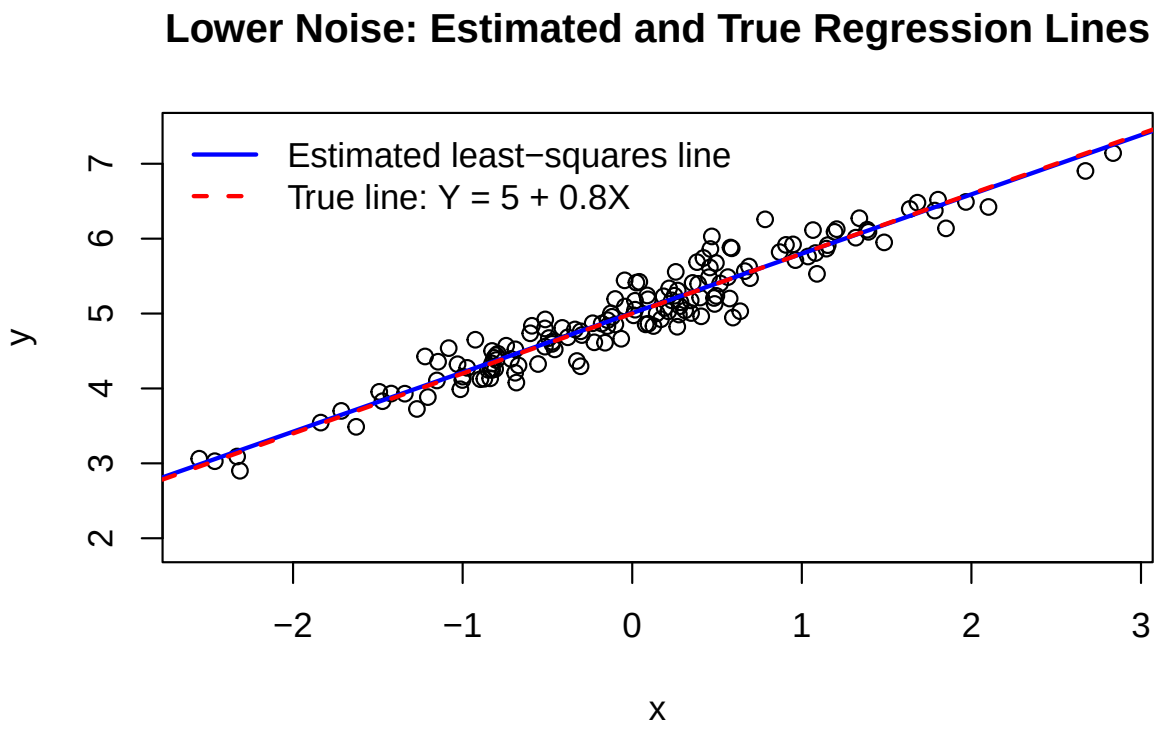
```
##
## Call:
## lm(formula = y_low_noise ~ x_low_noise)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.53099 -0.14229 -0.01291  0.12401  0.65113
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.00675    0.01797  278.56   <2e-16 ***
## x_low_noise  0.79206    0.01856   42.67   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2201 on 148 degrees of freedom
## Multiple R-squared:  0.9248, Adjusted R-squared:  0.9243
## F-statistic:  1821 on 1 and 148 DF,  p-value: < 2.2e-16
```

```r
cbind(True = c(5, 0.8),
      Original = coef(simple_regression_model),
      Lower_noise = coef(low_noise_model))
```

```
##             True  Original Lower_noise
## (Intercept)  5.0 5.0637427   5.0067535
## x            0.8 0.8535459   0.7920648
```

```r
plot(x_low_noise, y_low_noise,
     main = "Lower Noise: Estimated and True Regression Lines",
     xlab = "x", ylab = "y",
     xlim = range(x, x_low_noise), ylim = range(y, y_low_noise))
abline(low_noise_model, col = "blue", lty = 1, lwd = 2)
abline(a = 5, b = 0.8, col = "red", lty = 2, lwd = 2)
legend("topleft",
       legend = c("Estimated least-squares line", "True line: Y = 5 + 0.8X"),
       col = c("blue", "red"), lty = c(1, 2), lwd = 2,
       bty = "n")
```

Repeat (f): Overlay Estimated and True Regression Lines with a Legend



Effects on the Scatterplot, Parameter Estimates, and Overall Fit Reducing the noise results in the points in the scatterplots being more tightly clustered around the regression line. This causes the

linear pattern to become more clear. The estimated intercept changes from 5.0637427 to 5.0067535, and the estimated slope changes from 0.8535459 to 0.7920648. Both estimates from the lower noise model are closer to the true population parameters, $\beta_0 = 5$ and $\beta_1 = 0.8$. The overall fit also improves: R^2 increases from approximately 0.6770 to 0.9248, and the residual standard error decreases from approximately 0.5861 to 0.2201. This means that the lower noise model explains a larger proportion of the variation in y, and the observations lie closer to the fitted line.

(i) Original vs. Lower-Noise Models: 95% Confidence Intervals

```
original_ci <- confint(simple_regression_model, level = 0.95)
low_noise_ci <- confint(low_noise_model, level = 0.95)

ci_table <- data.frame(
  Dataset = rep(c("Original", "Lower noise"), each = 2),
  Coefficient = rep(c("Intercept (beta_0)", "Slope (beta_1)"), times = 2),
  Lower = c(original_ci[, 1], low_noise_ci[, 1]),
  Upper = c(original_ci[, 2], low_noise_ci[, 2]),
  row.names = NULL
)
knitr::kable(ci_table, digits = 5,
             col.names = c("Dataset", "Coefficient", "Lower 95% bound", "Upper 95% bound"),
             caption = "95% confidence intervals for the intercept and slope")
```

Confidence Intervals for the Intercept and Slope in Both Models

Table 1: 95% confidence intervals for the intercept and slope

Dataset	Coefficient	Lower 95% bound	Upper 95% bound
Original	Intercept (beta_0)	4.96879	5.15869
Original	Slope (beta_1)	0.75778	0.94931
Lower noise	Intercept (beta_0)	4.97124	5.04227
Lower noise	Slope (beta_1)	0.75539	0.82874

Comparison of Interval Widths and Theoretical Explanation The original intercept interval has width $5.15869 - 4.96879 = 0.18990$, while the lower-noise intercept interval has width $5.04227 - 4.97124 = 0.07103$. The original slope interval has width $0.94931 - 0.75778 = 0.19153$, while the lower-noise slope interval has width $0.82874 - 0.75539 = 0.07335$. This means reducing the noise makes both intervals substantially narrower, indicating more precise coefficient estimates.

Question 3: Predicting Crime Rates in the Boston Dataset

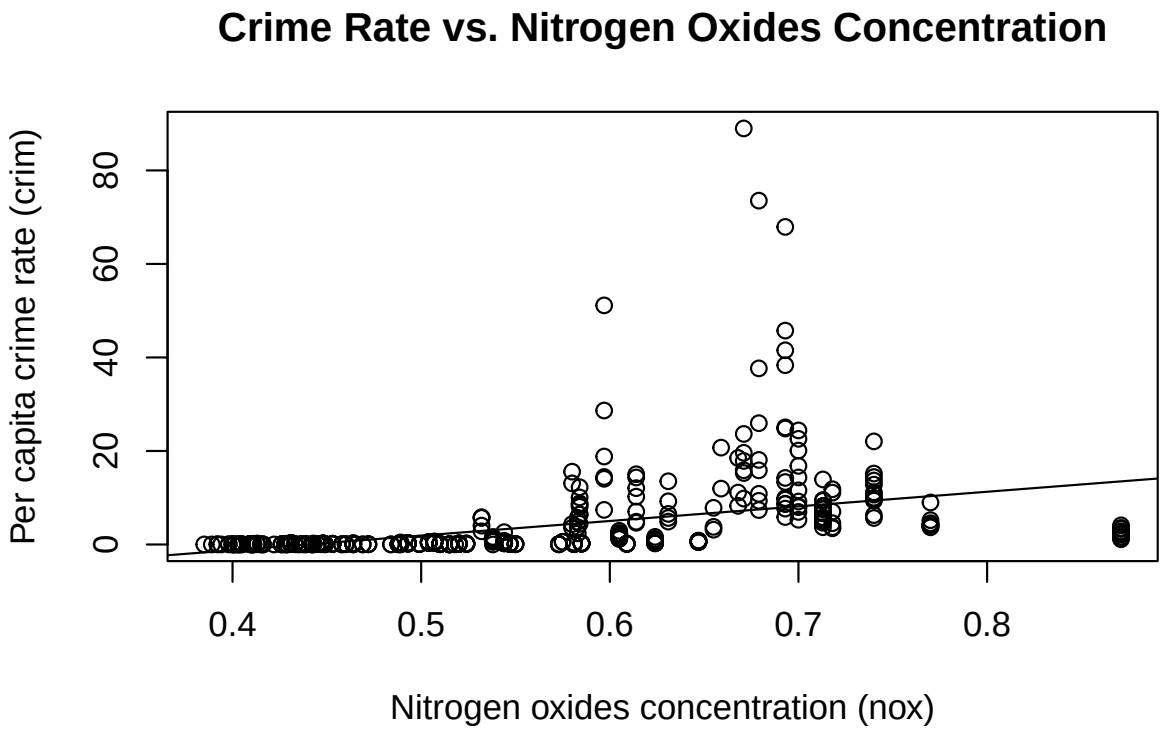
(a) Simple Linear Models: Results, Significance, and Scatterplots with Fitted Lines

```
boston_nox_model <- lm(crim ~ nox, data = Boston)
summary(boston_nox_model)
```

crim ~ nox

```
##
## Call:
## lm(formula = crim ~ nox, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.371  -2.738  -0.974   0.559  81.728
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -13.720      1.699  -8.073 5.08e-15 ***
## nox           31.249      2.999  10.419  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 504 degrees of freedom
## Multiple R-squared:  0.1772, Adjusted R-squared:  0.1756
## F-statistic: 108.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

```r
plot(crim ~ nox, data = Boston,
     main = "Crime Rate vs. Nitrogen Oxides Concentration",
     xlab = "Nitrogen oxides concentration (nox)",
     ylab = "Per capita crime rate (crim)")
abline(boston_nox_model)
```

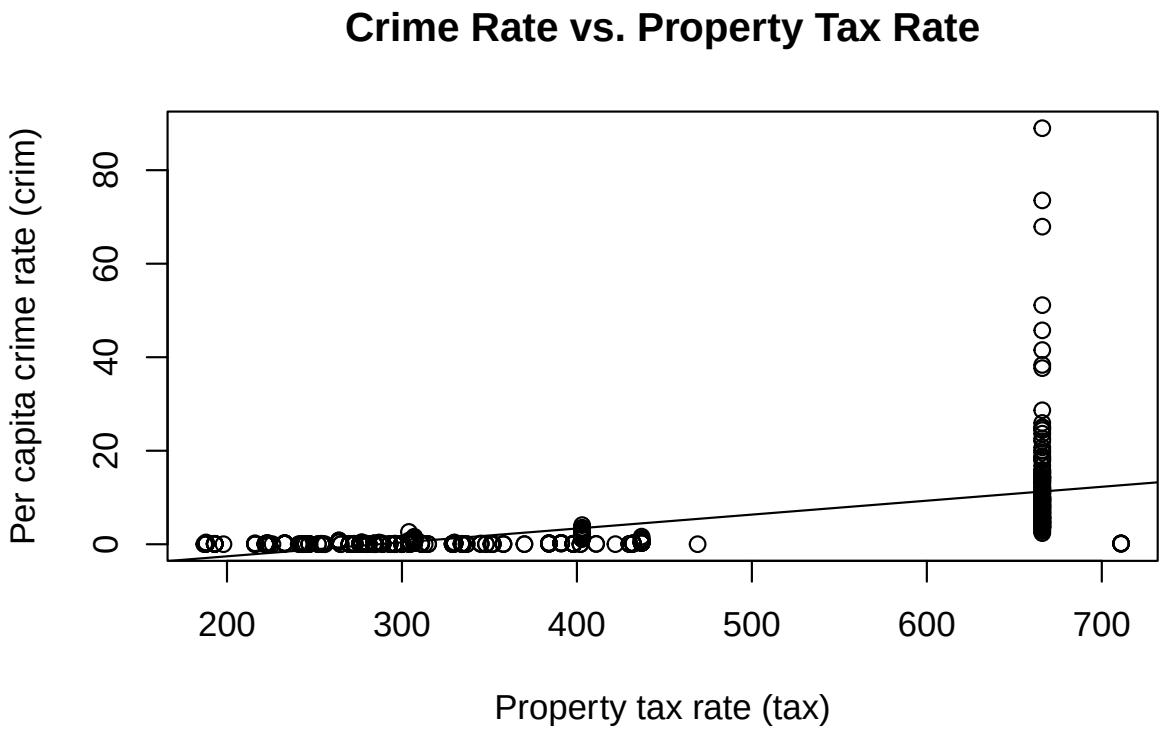


```r
boston_tax_model <- lm(crim ~ tax, data = Boston)
summary(boston_tax_model)
```

crim ~ tax

```
##
## Call:
## lm(formula = crim ~ tax, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.513  -2.738  -0.194   1.065  77.696
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.528369   0.815809  -10.45   <2e-16 ***
## tax          0.029742   0.001847   16.10   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.997 on 504 degrees of freedom
## Multiple R-squared:  0.3396, Adjusted R-squared:  0.3383
## F-statistic: 259.2 on 1 and 504 DF,  p-value: < 2.2e-16
```

```r
plot(crim ~ tax, data = Boston,
     main = "Crime Rate vs. Property Tax Rate",
     xlab = "Property tax rate (tax)",
     ylab = "Per capita crime rate (crim)")
abline(boston_tax_model)
```

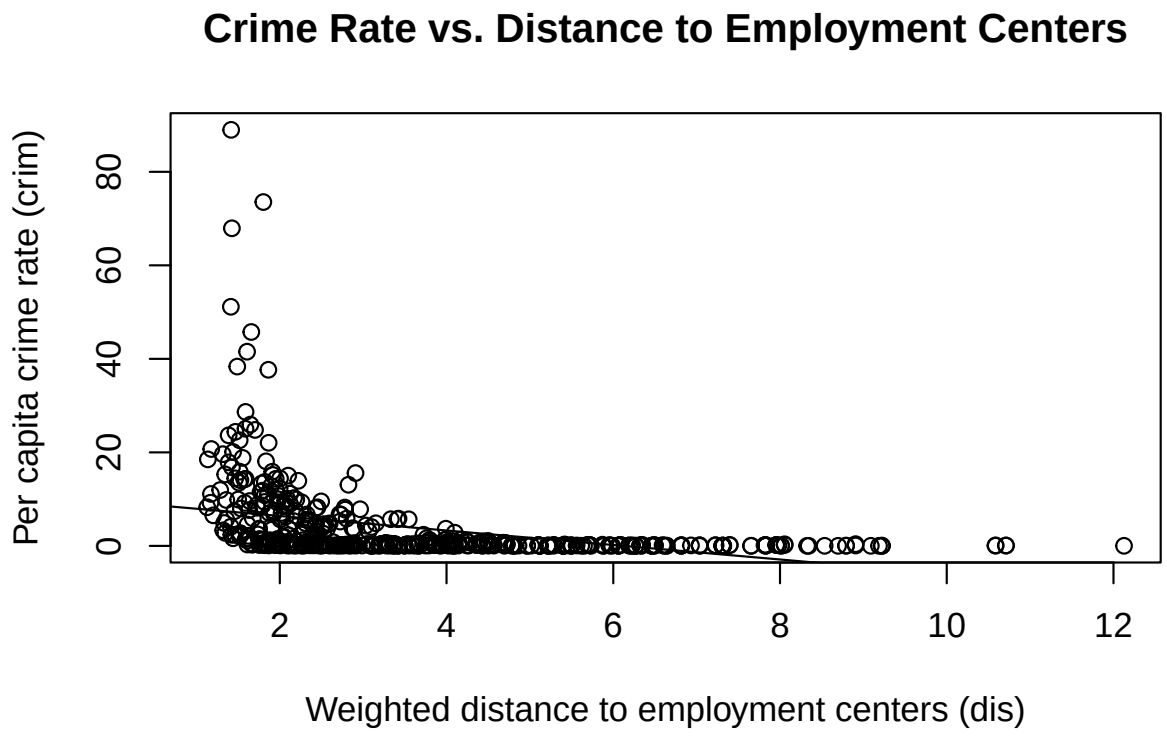



```
boston_dis_model <- lm(crim ~ dis, data = Boston)
summary(boston_dis_model)
```

crim ~ dis

```
##
## Call:
## lm(formula = crim ~ dis, data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.708 -4.134 -1.527  1.516 81.674
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.4993     0.7304  13.006   <2e-16 ***
## dis          -1.5509     0.1683  -9.213   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.965 on 504 degrees of freedom
## Multiple R-squared:  0.1441, Adjusted R-squared:  0.1425
## F-statistic: 84.89 on 1 and 504 DF,  p-value: < 2.2e-16
```

```r
plot(crim ~ dis, data = Boston,
     main = "Crime Rate vs. Distance to Employment Centers",
     xlab = "Weighted distance to employment centers (dis)",
     ylab = "Per capita crime rate (crim)")
abline(boston_dis_model)
```



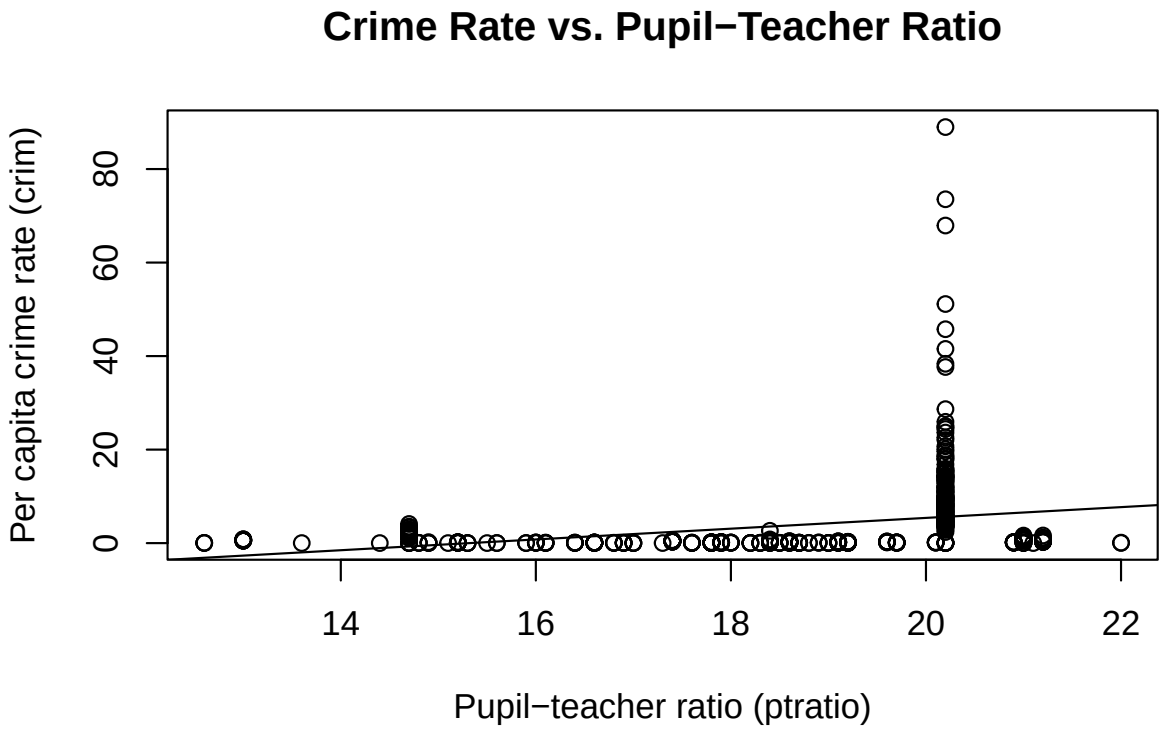

```r
boston_ptratio_model <- lm(crim ~ ptratio, data = Boston)
summary(boston_ptratio_model)
```

crim ~ ptratio

```
##
## Call:
## lm(formula = crim ~ ptratio, data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -7.654 -3.985 -1.912  1.825 83.353
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -17.6469     3.1473  -5.607 3.40e-08 ***
## ptratio       1.1520     0.1694   6.801 2.94e-11 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.24 on 504 degrees of freedom
## Multiple R-squared:  0.08407,    Adjusted R-squared:  0.08225
## F-statistic: 46.26 on 1 and 504 DF,  p-value: 2.943e-11
```

```r
plot(crim ~ ptratio, data = Boston,
     main = "Crime Rate vs. Pupil-Teacher Ratio",
     xlab = "Pupil-teacher ratio (ptratio)",
     ylab = "Per capita crime rate (crim)")
abline(boston_ptratio_model)
```



All four models show a statistically significant relationship between their associated predictor variable and the response variable crim. Each model produces a p-value less than 0.05, meaning that each predictor variable is statistically significant.

(b) Multiple Regression with All Available Predictors: Results and Significance at $\alpha = 0.05$

```r
boston_full <- lm(crim ~ ., data = Boston)
summary(boston_full)
```

```
##
## Call:
```

```
## lm(formula = crim ~ ., data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.534 -2.248 -0.348  1.087 73.923
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.7783938  7.0818258   1.946 0.052271 .
## zn           0.0457100  0.0187903   2.433 0.015344 *
## indus       -0.0583501  0.0836351  -0.698 0.485709
## chas        -0.8253776  1.1833963  -0.697 0.485841
## nox         -9.9575865  5.2898242  -1.882 0.060370 .
## rm           0.6289107  0.6070924   1.036 0.300738
## age         -0.0008483  0.0179482  -0.047 0.962323
## dis         -1.0122467  0.2824676  -3.584 0.000373 ***
## rad          0.6124653  0.0875358   6.997 8.59e-12 ***
## tax         -0.0037756  0.0051723  -0.730 0.465757
## ptratio     -0.3040728  0.1863598  -1.632 0.103393
## lstat        0.1388006  0.0757213   1.833 0.067398 .
## medv        -0.2200564  0.0598240  -3.678 0.000261 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.46 on 493 degrees of freedom
## Multiple R-squared:  0.4493, Adjusted R-squared:  0.4359
## F-statistic: 33.52 on 12 and 493 DF,  p-value: < 2.2e-16
```

At $\alpha = 0.05$, we reject the null hypothesis $H_0 : \beta_j = 0$ for the predictor variables zn ($p = 0.015344$), dis ($p = 0.000373$), rad ($p = 8.59 \times 10^{-12}$), and medv ($p = 0.000261$). Each has a p-value below 0.05, which means that they are not statistically significant. We fail to reject the null hypothesis for the remaining predictors.

(c) Simple vs. Multiple Regression Results

In the simple linear regression models, all four predictor variables nox, tax, dis, and ptratio were found to be statistically significant. This changes significantly in the multiple linear regression model. The predictor variables nox, tax, and ptratio change signs and are no longer statistically significant at $\alpha = 0.05$. The predictor variable dis remains negative and is still statistically significant, but it does decrease in magnitude from -1.5509 to -1.0122467. These differences can occur in the multiple linear regression model due to factors such as confounding and collinearity. Confounding occurs when a predictor may appear to be related to the response variable but is actually more closely related to other predictor variables that themselves are more closely related to the response variable. Collinearity occurs when the predictor variables are correlated with each other making it difficult for the multiple linear regression model to separate the individual effects. These effects can cause the changes in magnitude, sign, and statistical significance in the predictor variables.

(d) Cubic Polynomial Models for Each Predictor: Nonlinear Associations

```
predictors <- setdiff(names(Boston), "crim")
cubic_models <- setNames(vector("list", length(predictors)), predictors)
```

```r
nonlinear_results <- data.frame(
  Predictor = predictors,
  Quadratic_p = NA_real_,
  Cubic_p = NA_real_
)

for (j in seq_along(predictors)) {
  predictor <- predictors[j]
  quadratic_term <- paste0("I(", predictor, "^2)")
  cubic_term <- paste0("I(", predictor, "^3)")
  model_formula <- reformulate(
    c(predictor, quadratic_term, cubic_term), response = "crim"
  )
  cubic_models[[predictor]] <- lm(model_formula, data = Boston)
  coefficients <- coef(summary(cubic_models[[predictor]]))

  # Aliased terms (such as chas^2 and chas^3) have no separate p-values.
  nonlinear_results$Quadratic_p[j] <- coefficients[
    match(quadratic_term, rownames(coefficients)), "Pr(>|t|)"
  ]
  nonlinear_results$Cubic_p[j] <- coefficients[
    match(cubic_term, rownames(coefficients)), "Pr(>|t|)"
  ]
}

p_value_table <- nonlinear_results
p_value_table$Quadratic_p <- format.pval(nonlinear_results$Quadratic_p, digits = 4)
p_value_table$Cubic_p <- format.pval(nonlinear_results$Cubic_p, digits = 4)
knitr::kable(
  p_value_table,
  col.names = c("Predictor", "Quadratic term p-value", "Cubic term p-value"),
  caption = "Tests of nonlinear terms in separate cubic regression models"
)
```

Model Fits and Hypothesis Tests for the Quadratic and Cubic Terms

Table 2: Tests of nonlinear terms in separate cubic regression models

Predictor	Quadratic term p-value	Cubic term p-value
zn	0.09375	0.229539
indus	3.420e-10	1.196e-12
chas	NA	NA
nox	6.811e-15	6.961e-16
rm	0.36411	0.508575
age	0.04738	0.006680
dis	4.941e-12	1.089e-08
rad	0.61301	0.482314
tax	0.13747	0.243851
ptratio	0.00412	0.006301
lstat	0.06459	0.129891
medv	< 2.2e-16	1.047e-12

Evidence and Conclusions Based on p-Values For each predictor, we test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$ and $H_0 : \beta_3 = 0$ against $H_1 : \beta_3 \neq 0$. We continue to use $\alpha = 0.05$.

There is evidence of nonlinear association for indus, nox, age, dis, ptratio, and medv: both their quadratic and cubic terms have p-values below 0.05. For zn, rm, rad, tax, and lstat, neither the quadratic nor the cubic term has a p-value below 0.05. These individual tests do not provide sufficient evidence of nonlinear association at this significance level.